

Project Submission Details

Project Title: Prarup: A Submission Compliance Checker for Research Papers

Group Details

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1. Description

Conference papers are often rejected before anyone reads them, for formatting faults the author cannot see. The most common fault sits inside the figures. Plotting libraries such as matplotlib, MATLAB and R export charts whose fonts are either not embedded in the file or stored as bitmaps, and IEEE rejects both. Nothing in the PDF shows this, nothing appears in the compilation log, and the author finds out on the day of the deadline, from a validator that replies only with a failure.

Prarup is an offline desktop application, built in Python, that checks a paper before it is submitted. It takes a manuscript in Word or LaTeX format, fits it to the target conference template, compiles it, and then examines the resulting PDF for the faults that actually cause rejection. Every check is ordinary code with a fixed answer, not a judgement by a model. Faults that can be corrected safely are corrected, and the file is checked again afterwards to confirm the correction worked. Faults that cannot be corrected are reported as such, with the change the author needs to make. Nothing is uploaded anywhere; the paper stays on the author's computer.

2. Objectives

- **Manuscript Conversion:** To read papers written in Word or LaTeX and convert them into the required conference format, keeping figures, tables and equations intact.
- **Live Compilation:** To build the paper and show the finished PDF as the author works, so formatting problems are visible immediately rather than at the end.
- **Compliance Checking:** To inspect the compiled PDF for embedded fonts, bitmap fonts, archival format, page limits and document permissions, measuring each one directly from the file.
- **Automatic Correction:** To repair the faults that have a safe mechanical fix, verify the repair by running the checks a second time, and clearly separate these from faults that need the author to redo a figure.
- **Offline Operation:** To run the whole process on the user's own machine, so unpublished work never has to be sent to an online service to be checked.

3. Outcomes

- **Working Desktop Application:** A single program where the paper is opened, formatted, compiled and checked, with the document and the list of problems shown side by side.
- **Problems Found Before Submission:** The verdict the publisher's validator gives at the end, moved to a point where the author can still act on it, with the reason attached.
- **Reusable Rule Set:** Conference requirements kept in editable rule files, so supporting another conference means adding a file rather than changing the program.
- **Submission Package:** An archive holding the source, the figures and a short report stating which checks the paper passed, ready to upload.